Constructing Newton motion: refinement, continuation and regularity

Maintained result. For a globally Lipschitz acceleration, arbitrary positive partitions and independently mixed kick–drift orders construct the same unique initial-value motion, with an explicit error proportional to the mesh. A conservative logarithmic force family shows the exact limit of that sufficient hypothesis: an unbounded slope still permits unique motion when $0 < \beta \leq 2$; finite-time arrival and waiting branches occur for $\beta > 2$. Local action tends to zero in both regimes. These are classical existence and uniqueness statements, with no positive action premise.

The polygon proof uses standard stability, consistency and Gronwall arguments. The arrival test is a direct energy calculation, not the general first-order Osgood criterion. Teschl [@TeschlODE2012, §2.7, pp.54–56 and Problem 2.25, p.58; passages read] supplies the Euler compactness background; López Pouso [@LopezPouso2005; metadata and abstract read] treats a broader second-order problem whose full theorem is not used here. See the source companion. Internal Terra/Sol checks of the written arguments were model reviews, not independent peer review or novelty evidence.

1. Start with a finite Newton polygon

Let $m > 0$ be the mass and $f(q) = F(q)/m$ the acceleration. The desired motion would satisfy

$$\dot{q} = v, \qquad \dot{v} = f(q), \qquad (q(0), v(0)) = (q_0, v_0). \tag{1}$$

For now we do not assume that a solution exists. Given a positive time step τ, define a drift and a force kick by

$$D_\tau(q, v) = (q + \tau v, v), \qquad K_\tau(q, v) = (q, v + \tau f(q)). \tag{2}$$

A step can kick first, $D_\tau K_\tau$, or drift first, $K_\tau D_\tau$. The first choice uses the force at the old position; the second uses the force at the new position. Both give a finite piecewise straight trajectory with impulses. Neither is assumed to agree exactly with the unknown continuous motion.

Fix $T > 0$, take any finite positive partition of $[0, T]$, choose either ordering in each cell, and compose the maps chronologically. There is no equal-step or nesting requirement. When a time is inserted, recompute the later states from the fixed initial state and the prescribed force rule.

Existence of the limit requires two kinds of estimates. Stability bounds how differences grow. Consistency bounds the error in the finite force balance. The two together will construct the ODE solution.

2. Lipschitz control supplies stability

Assume for this construction that

$$|f(q) - f(Q)| \leq \Lambda |q - Q| \quad (q, Q \in \mathbb{R}), \qquad \Lambda \geq 0. \tag{3}$$

The quantity Λ has units of inverse time squared. Choose a reference frequency $\rho > 0$ and use the length-valued phase-space norm

$$\|(q, v)\|_\rho = |q| + \frac{|v|}{\rho}, \qquad \alpha = \max(\rho, \Lambda/\rho), \quad \beta = \rho + \Lambda/\rho, \quad b = |f(0)|/\rho. \tag{4}$$

There is no action constant in these definitions. Directly from (2),

$$\mathrm{Lip}(D_\tau) \leq 1 + \rho\tau, \qquad \mathrm{Lip}(K_\tau) \leq 1 + \Lambda\tau/\rho. \tag{5}$$

Thus either cell map has Lipschitz constant at most $e^{\beta\tau}$. The constants multiply under composition, and their exponents add. Every finite polygon therefore has the same upper amplification $e^{\beta T}$, independently of the number and placement of vertices.

We also need a bound on the states themselves, because consistency involves the force along the polygon. For either cell map M_τ,

$$\|M_\tau z\|_\rho \le e^{\beta\tau}\|z\|_\rho + \tau e^{\beta\tau} b. \tag{6}$$

For a kick-first cell, the inhomogeneous term is actually at most $\tau(1+\rho\tau)b$; for a drift-first cell it is τb. Iterating (6) gives the common vertex bound

$$R = e^{\beta T}(\|z_0\|_\rho + Tb). \tag{7}$$

In particular $|q_j| \le R$, $|v_j| \le \rho R$, and $|f(q_j)| \le A := |f(0)| + \Lambda R$. A very irregular partition has not created a new force or stability constant.

3. The polygons construct the ODE solution

Linearly interpolate both vertex coordinates to obtain $z_\pi(t) = (q_\pi(t), v_\pi(t))$. The displayed velocity is continuous; it differs from the physical constant velocity along each straight polygon segment by at most that cell's kick. This distinction disappears with the mesh.

Let $\delta = |\pi|$ be the maximum cell duration. In a kick-first cell, with elapsed time $s \in [0, \tau]$,

$$\dot q_\pi - v_\pi = (\tau - s) f(q_j), \qquad \dot v_\pi - f(q_\pi) = f(q_j) - f(q_\pi). \tag{8}$$

In a drift-first cell the analogous expressions are $-sf(q_{j+1})$ and $f(q_{j+1}) - f(q_\pi)$. The bounds in (7) give, for almost every time,

$$\|\dot z_\pi - \mathcal{F}(z_\pi)\|_\rho \le C\delta, \quad \mathcal{F}(q,v) = (v, f(q)), \quad C = |f(0)| + 2\Lambda R. \tag{9}$$

The vector field $\mathcal{F}$ is Lipschitz with constant α in the norm (4). Two polygons with the same initial state consequently satisfy

$$\sup_{t \le T} \|z_\pi(t) - z_\sigma(t)\|_\rho \le C \frac{e^{\alpha T} - 1}{\alpha} (|\pi| + |\sigma|). \tag{10}$$

To verify this estimate, integrate the two residual equations (9), subtract, and apply the elementary integral Gronwall inequality. That inequality follows by integrating the differential inequality for an upper bound after multiplying by $e^{-\alpha t}$.

Completeness of continuous path space now constructs a uniform limit $z(t)$. The residual integral tends to zero; the Lipschitz bound passes $\mathcal{F}(z_\pi)$ to $\mathcal{F}(z)$ under the time integral. Hence

$$z(t) = z_0 + \int_0^t \mathcal{F}(z(s))\, ds. \tag{11}$$

The limit is differentiable and solves (1). Applying the same difference estimate with zero residual proves uniqueness. Letting one mesh tend to zero in (10) gives a quantitative error for any other partition. Both kick orders, mixed arbitrarily, lead to the same solution. Since T was arbitrary, the global Lipschitz hypothesis constructs forward motion for every finite time.

This is the first analogy with Balaban: one proves bounds for each finite description before appealing to a limiting object. The limit follows from estimates uniform in the refinement. Nothing in this argument requires choosing $\hbar > 0$.

4. Local continuation and velocity-dependent forces

For locally Lipschitz acceleration, extend it from a bounded region and apply the construction until the trajectory leaves that region. Continuation then requires bounds preventing escape from every compact phase-space region.

For $V \in C^1$ with locally Lipschitz ∇V, (1) has a unique local solution, by the usual Cauchy–Lipschitz theorem. Along it,

$$E(q,v) = \tfrac{1}{2}m|v|^2 + V(q), \qquad \frac{d}{dt}E = 0.$$

If V is bounded below, (2) bounds $|v|$. On any finite interval $|q(t)| \le |q_0| + |t| \sup |v|$, so (q,v) remains in a compact set. The local continuation theorem then gives a unique solution for all positive and negative times. Coercivity of V is a stronger condition than is needed for this finite-time argument.

Smoothness alone does not guarantee global existence. For $m = 1$, $V(q) = -q^4/4$, $q_0 > 0$ and $v_0 = q_0^2/\sqrt{2}$, the zero-energy solution

$$q(t) = \frac{q_0}{1 - q_0 t/\sqrt{2}}$$

escapes to infinity at $t = \sqrt{2}/q_0$. A force singularity can also stop a classical solution: for example, a radial attractive Kepler trajectory can reach $r = 0$ in finite time. A rule for continuing through the singular point is further data.

In three dimensions the Lorentz force, with a static scalar potential and a static magnetic field, gives

$$\dot{q} = v, \qquad m\dot{v} = -\nabla V(q) + e\, v \times B(q).$$

It is not a force depending on position alone. If ∇V and B are defined and locally Lipschitz on all of $\mathbb{R}^3$, the full phase-space vector field is locally Lipschitz, so local existence and uniqueness still hold. Moreover $v \cdot (v \times B) = 0$, hence the conservative energy is conserved. If V is bounded below, the same speed and position bounds as above prove global existence and uniqueness, even if B is not globally bounded.

For instance, if V is constant and $B = B_0\hat{z}$, the field rotates the transverse velocity with cyclotron frequency eB_0/m and leaves the longitudinal velocity constant. A velocity-dependent force is thus not an exception to the ODE theory; its dependence must be included in the phase-space vector field being estimated.

If $B = \nabla \times A$ on $\mathbb{R}^3$, the quantum Hamiltonian is

$$H_{\hbar,A} = \frac{1}{2m}(-i\hbar\nabla - eA(q))^2 + V(q).$$

The kinetic momenta generally do not commute: $[\pi_i, \pi_j] = i\hbar e(\partial_i A_j - \partial_j A_i)$. A positive scalar Brownian-bridge density does not directly cover this magnetic problem: a magnetic Euclidean representation contains a phase. The coherent-packet theorem treats the scalar Hamiltonian; it does not prove this magnetic extension.

An endpoint problem fixes $q(0)$ and $q(T)$ rather than initial velocity. It has a separate possible singularity. For the harmonic oscillator the full phase-space derivative is

$$\begin{pmatrix} \cos\omega T & \sin(\omega T)/\omega \\ -\omega \sin(\omega T) & \cos\omega T \end{pmatrix}, \qquad \det = 1.$$

The position response to initial velocity is $\sin(\omega T)/\omega$. It vanishes at a conjugate time while the full phase-space map stays invertible. The corresponding endpoint action Hessian can fail to invert even though the initial-value motion remains unique. The time-blocking note gives a finite-polygon determinant that approaches this singularity under refinement.

Therefore a failure of one position-only generating function is not, by itself, a failure of Newton's dynamics. A physical completion argument has to establish which extra information the complete theory needs and why it cannot be retained classically.

5. The conservative arrival criterion

Near zero let a be continuous, odd, positive on $(0, r_0)$, zero at zero and smooth away from zero. Consider

$$\ddot{q} = a(q), \qquad W(q) = \int_0^q a(s)\,ds \quad (q \geq 0), \qquad V(q) = -mW(|q|), \quad m > 0. \tag{L1}$$

Assume a bounded continuous global extension of the force, smooth away from zero, by cutting off the potential outside a neighborhood of zero. The bounded-cusp argument, section 1, then gives global existence and bounds preventing escape. The issue remaining here is uniqueness, not existence.

Conservation of $E = m\dot{q}^2/2 + V(q)$ follows by the chain rule since $V \in C^1$. Any solution starting at rest at zero has $E = 0$. On a positive departing segment its velocity must be

$$\dot{q} = \sqrt{2W(q)}, \qquad \mathcal{T}(q) = \int_0^q \frac{ds}{\sqrt{2W(s)}}. \tag{L2}$$

Arrival criterion. If $\mathcal{T}(q) = \infty$ for one, hence every small $q > 0$, rest is the unique local solution from $(0, 0)$. If $\mathcal{T}(q) < \infty$, the inverse $Q = \mathcal{T}^{-1}$ constructs departures $q(t) = \pm Q(t - t_0)$ after every waiting time t_0.

For the first implication, a nonconstant solution has an excursion of one sign starting from zero. Its velocity cannot vanish inside that excursion because $W > 0$, and separation of variables would give a finite value for the divergent integral (L2). For the converse, $Q' = \sqrt{2W(Q)}$ and $Q'' = a(Q)$. Both tend to zero at departure, so adjoining rest gives a C^2 Newton trajectory. Reversing the two departures supplies two different incoming full states that merge at zero in finite time, exactly as in the cusp note.

For $a(q) = cq^\alpha$ this test reproduces the familiar $\alpha < 1$ nonuniqueness threshold: its integrand is proportional to $q^{-(1+\alpha)/2}$. The logarithmic family below tests the more delicate border where no power-law Lipschitz constant exists.

Exact logarithmic threshold

Fix $\Omega, \ell > 0$, $\beta > 0$ and put $r = \beta/2$. For sufficiently small $q > 0$, with $L = \log(\ell/q) > \max\{1, r\}$, set

$$W(q) = \frac{\Omega^2}{2} q^2 L^\beta, \qquad a(q) = \Omega^2 q(L^\beta - rL^{\beta-1}). \tag{L3}$$

Extend a oddly near zero and the potential with a smooth cutoff farther out. This gives a bounded C^1 potential and continuous bounded force; $a(0) = 0$ since $qL^\beta \to 0$. Its derivative on the positive side is

$$a'(q) = \Omega^2 \left[L^\beta - \frac{3\beta}{2} L^{\beta-1} + \frac{\beta(\beta-1)}{2} L^{\beta-2} \right], \tag{L4}$$

which diverges for every $\beta > 0$. Thus none of these forces is locally Lipschitz at zero.

Nevertheless the energy-zero speed and arrival integral are exact:

$$\dot{q} = \Omega q L^r, \qquad \mathcal{T}(q) = \frac{1}{\Omega} \int_L^\infty u^{-r}\,du. \tag{L5}$$

Consequently rest is unique for $\beta \leq 2$. For $\beta > 2$,

$$\mathcal{T}(q) = \frac{L^{1-r}}{\Omega(r-1)}, \qquad Q(t) = \ell \exp\left(-[\Omega(r-1)t]^{-1/(r-1)} \right). \tag{L6}$$

The formula for Q is valid until the departure leaves the region where (L3) is used. It is flat to every order at $t = 0$, so an exactly resting body and one that leaves after a delay satisfy the same Newton equation and initial data.

For $\beta \le 2$ the *whole* extended IVP is unique. Away from zero this is ordinary local Lipschitz uniqueness. At zero with nonzero initial velocity, the conserved-energy first-order equation has a nonzero square root and a C^1 right-hand side near the crossing, so it gives unique passage. At zero velocity the arrival criterion applies. Bounded force then gives a global reversible point flow. Its dependence on initial state is continuous: common finite-time bounds give compactness for any converging initial sequence; every subsequential limit solves the same unique IVP. Thus all subsequences have the same limit. A divergent local slope has not destroyed reversible mechanics.

At the borderline $\beta = 2$, the outgoing energy-zero solution from $q_0 > 0$ is particularly informative:

$$q(t) = \ell \left(\frac{q_0}{\ell}\right)^{e^{-\Omega t}}, \qquad \dot{q}(0) = \Omega q_0 \log(\ell/q_0). \tag{L7}$$

While it remains in the logarithmic region, it stays continuous as $q_0 \downarrow 0$ at every fixed finite time, although its position sensitivity diverges. This is a genuine example of unique stable motion without a Lipschitz dependence bound at rest.

The general first-order Osgood sufficient test applied directly to a modulus of size $s[\log(1/s)]^\beta$ reaches only $\beta \le 1$. The stronger range $1 < \beta \le 2$ above uses the conservative Newton structure and energy; failure of that first-order sufficient test is not a proof of nonuniqueness.

Crossing times and arbitrary-mesh polygons

Let $q_n = \ell e^{-n}$ with n sufficiently large. Along the zero-energy trajectory, the time for the annulus $q_{n+1} < q < q_n$ is

$$\tau_n = \frac{1}{\Omega} \int_n^{n+1} u^{-r}\, du, \qquad \tau_n \sim \frac{1}{\Omega n^r}. \tag{L8}$$

On this annulus the local acceleration slope bound is of order $\Lambda_n = \Omega^2 n^\beta$. The short-cell scale $\Lambda_n^{-1/2}$ is therefore of the same order as τ_n. One can subdivide an annulus crossing into a fixed number of shorter steps to obtain a small local Lipschitz amplification parameter. There is no uniform slope bound across all annuli.

The issue is whether infinitely many controlled annulus crossings can fit into finite physical time:

$$\sum_{n \ge N} \tau_n \begin{cases} = \infty, & \beta \le 2, \\ < \infty, & \beta > 2. \end{cases} \tag{L9}$$

For $\beta \le 2$, a zero-energy trajectory cannot reach the infinite-depth boundary at zero in finite time. For $\beta > 2$, perfectly regular motion on each finite collection of annuli reaches that boundary in finite time, and the branch datum is missing there. Local estimates at every finite scale do not by themselves control their infinite composition. This is the useful multiscale connection with the Balaban lesson; it is not a transfer of his field-theory proof.

For the unique regime, arbitrary-mesh Euler polygons, and the lesson's mixed kick–drift polygons, converge uniformly on every fixed finite time interval. Bounded force gives common position and momentum bounds; uniform continuity gives a vanishing integral residual; compactness supplies solution subsequences; uniqueness makes the whole family converge. No $O(\text{mesh})$ rate is asserted here.

In the nonunique regime, exact zero initial data make these deterministic polygons remain at rest on every partition. However, preparations $q_0 \downarrow 0$, $\dot{q}_0 = \Omega q_0 L_0^r \downarrow 0$ have

$$L(t) = \Big[L_0^{1-r} + \Omega(r-1)t\Big]^{-1/(r-1)} \longrightarrow [\Omega(r-1)t]^{-1/(r-1)}. \tag{L10}$$

Thus they converge to the nonzero departure (L6) at fixed small $t > 0$. For each positive q_0, choose a fine enough polygon to approximate that regular trajectory; a diagonal sequence then has vanishing mesh and vanishing initial error but a departing limit. Refinement is not

jointly stable with preparation refinement at zero. This is a stronger failure than an unbounded numerical error estimate, while keeping the exact-zero polygon's resting limit explicit.

Vanishing local action and a quantum residual

The positive outgoing momentum is $p(q) = m\Omega q L^r$. Its canonical action from zero to q is

$$\mathcal{J}(q) = \int_0^q p(s)\, ds = m\Omega\ell^2 \int_L^\infty e^{-2u} u^r\, du. \tag{L11}$$

For $L > r/2$, elementary bounds give

$$\frac{m\Omega}{2} q^2 L^r \le \mathcal{J}(q) \le \frac{m\Omega q^2 L^r}{2 - r/L}. \tag{L12}$$

The lower bound uses $u^r \ge L^r$; the upper bound uses $(L+v)^r \le L^r e^{rv/L}$ after writing $u = L + v$. Therefore $\mathcal{J}(q) \sim (m\Omega/2) q^2 L^r \to 0$ for every $\beta > 0$. The temporal threshold (L9) creates no positive action threshold.

There is still a concrete quantum comparison. In the local region $V = -p(q)^2/(2m)$, take the stationary energy-zero test functions

$$\psi_\pm(q) = p(q)^{-1/2} \exp\left(\pm \frac{i}{\hbar} \mathcal{J}(q)\right), \qquad H_\hbar = -\frac{\hbar^2}{2m} \frac{d^2}{dq^2} + V(q). \tag{L13}$$

Their eikonal and first-derivative transport terms cancel exactly. Writing $b(q) = p(q)^{-1/2}$ gives $H_\hbar \psi_\pm = -(\hbar^2/2m) b'' e^{\pm i\mathcal{J}/\hbar}$ and the exact local ratio

$$\frac{|H_\hbar \psi_\pm|}{|V\psi_\pm|} = \left[\frac{\hbar}{m\Omega q^2 L^r}\right]^2 \left|\frac{3}{4} - \frac{r}{L} + \frac{r(r+2)}{4L^2}\right| \sim \frac{3\hbar^2}{16\mathcal{J}(q)^2}. \tag{L14}$$

For example, the differentiation behind this identity is $p'/p = (1 - r/L)/q$ and $p''/p = -r[1 - (r-1)/L]/(q^2 L)$; then $b''/b = 3(p'/p)^2/4 - p''/(2p)$. Equation (L14) is a local residual identity, not an error bound for a normalized state or the full propagator.

At fixed $\hbar > 0$, action comparable with $\hbar$ makes the local residual of this trajectory ansatz order one even in the classically unique regime. Every sufficiently small positive $\hbar$ defines a radius $q_\hbar$ by $\mathcal{J}(q_\hbar) = \hbar$, and that radius tends to zero as $\hbar \to 0$. The bounded-potential Hamiltonian has global unitary dynamics at each positive $\hbar$ by the cusp note's held Kato–Rellich input. These facts neither require $\hbar = 0$ to be a member of the same state theory nor select a positive numerical lower bound across the family. No relativistic ingredient or c is used.

6. Consequence for STATE

Initial-value construction, continuation, endpoint caustics and finite-time mergers have different hypotheses. Unbounded force slope alone establishes no merger and no action floor. The cusp preparation theorem uses an actual full-state merger and additional preparation laws. The separate endpoint blocking argument retains inverse Hessians, normalization and source response; the nonlinear cell controls a supplied positive weight. Neither operation follows solely from constructing a classical orbit.